



# The ProbLemma Channel Recurrence Relations

This document links all the recurrence relations evaluated on the ProbLemma channel to the episodes in which these evaluations were carried out.



In addition, all the above episodes are collected in [the “Recurrence Relations Are Us” play list](#).

## 1. Season 3 Episode 41 (S3M22): How Generating Functions Can Solve Linear Non-homogeneous Recurrence Relations

$$t_n = 6t_{n-1} - t_{n-2} - 4, \quad t_0 = 2, t_1 = 4$$

An analytic representation of a generating ordinary power series function for the above recurrence relation is:

$$G(x) = \frac{1}{1-x} - \frac{3+2\sqrt{2}}{2(x-3-2\sqrt{2})} - \frac{3-2\sqrt{2}}{2(x-3+2\sqrt{2})}$$

and its power series decomposition is:

$$G(x) = \sum_{n=0}^{+\infty} \left[ 1 + \frac{1}{2} (3-2\sqrt{2})^n + \frac{1}{2} (3+2\sqrt{2})^n \right] x^n$$

As such:

$$t(n) = 1 + \frac{1}{2} (3-2\sqrt{2})^n + \frac{1}{2} (3+2\sqrt{2})^n$$

with the first few terms:

$$t_n = \{2, 4, 18, 100, 578, 3364, \dots\}$$

This relation is also solved purely geometrically in [the “Inversive Geometry” course](#) in following episode: [IGRP Episode 20 Prctice 2. Problem 21. Inversion Meets A Recurrence Relation](#).

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## 2. Season 3 Episode 42 (S3M23): Chords of the ring or varying textures of four solutions

$$c_{n+1} = c_n + n, \quad c_0 = c_1 = 0$$

An analytic representation of a generating ordinary power series function for the above recurrence relation is:

$$g(x) = \frac{x^2}{(1-x)^3}$$

and its power series decomposition is:

$$g(x) = \sum_{n=0}^{+\infty} \frac{n(n-1)}{2} x^n$$

implying that:

$$c_n = \frac{n(n-1)}{2} = \{0, 0, 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, \dots\}$$

The above sequence  $\{c_n\}$  was generated as an answer to the question about the number of chords formed by  $n$  distinct points located on the circumference of a circle.

The next sequence,  $\{r_n\}$ , describes an answer to a similar question about the number of planar regions defined by  $n$  distinct straight lines in a general position:

$$r_{n+1} = r_n + n + 1, \quad r_0 = 1$$

An analytic representation of a generating ordinary power series function for the above recurrence relation is:

$$f(x) = \frac{1}{1-x} + \frac{x}{(1-x)^3}$$

and its power series decomposition is:

$$f(x) = \sum_{n=0}^{+\infty} \left[ 1 + \frac{n(n+1)}{2} \right] x^n$$

implying that:

$$r_n = 1 + \frac{n(n+1)}{2} = \{1, 2, 4, 7, 11, 16, 22, 29, 37, 46, 56, 67, \dots\}$$

The next sequence,  $\{M_n\}$ , describes a recursive solution of a classical Towers Of Hanoi puzzle with three rods,  $n$  discs and no extra restrictions beyond the traditional ones.

As such:

$$M_n = 2M_{n-1} + 1, \quad M_0 = 0$$

An analytic representation of a generating ordinary power series function for the above recurrence relation is:

$$g(x) = \frac{x}{(1-2x)(1-x)} = \frac{1}{1-2x} - \frac{1}{1-x}$$

and its power series decomposition is:

$$g(x) = \sum_{n=0}^{+\infty} [2^n - 1] x^n$$

implying that:

$$M(n) = 2^n - 1 = \{0, 1, 3, 7, 15, 31, 63, 127, 255, 511, 1023, 2047, 4095, \dots\}$$

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### 3. Season 3 Episode 66 (S3M35): OPSGFs Versus Fibonacci Numbers Or Mathematics Is A Process

$$F_{n+1} = F_n + F_{n-1}, \quad F_0 = 0, F_1 = 1$$

$$g(x) = \frac{x}{1-x-x^2} = \frac{1}{\varphi - \psi} \left( \frac{1}{1-x\varphi} - \frac{1}{1-x\psi} \right)$$

where:

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2}$$

and:

$$g(x) = \sum_{n=0}^{+\infty} \frac{\varphi^n - \psi^n}{\sqrt{5}} x^n$$

implying that:

$$F(n) = \frac{1}{\sqrt{5}}(\varphi^n - \psi^n) = \{0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, \dots\}$$

4. Season X Episode Y (SxMy): Lorem Ipsum

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